Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: FindYourDent
b. Team leader name: Ahmad Wali Radhi
c. Problem Statement: Survei Kesehatan Indonesia 2023: 56.9% of Indonesians aged 3 and over had a dental or oral problem in the past year, yet only 11.2% of them were treated by dental health workers. A Patient cannot tell how urgent it is, what it will cost, whether BPJS pays, or where to go.

Sources: SKI 2023 Dalam Angka, BKPK Kemenkes, Tabel 9.5 p. 324 (56.9%) and Tabel 9.7 p. 327 (57.0%); BKPK SKI 2023 oral health factsheet, Gambar 3 p. 2 (11.2%).

Brief about the idea

A Patient describes a toothache in their own words and leaves knowing how urgent it is, what it costs, and where to go

THE GAP · SKI 2023

56.9% of Indonesians aged 3+ had a dental or oral problem in the past year

11.2% of them were treated by dental health workers

57.0% of people not treated by a dental health worker said they felt no need

WHAT FINDYOURDENT DOES

1. **Describe**
   Type or speak the complaint in Indonesian, a regional language or English. The complaint is never stored.

2. **Guidance**
   Urgency tier, likely category, visit journey, cost estimate with why prices vary, and BPJS coverage. Red flags go straight to “Ke IGD sekarang”.

3. **Shortlist**
   Nearby Providers that can treat this complaint, with Testimonies and price signals.

4. **Testimony**
   After the visit the Patient tells others. Gemini screens it against fixed rules; a proof of visit earns “Kunjungan terbukti”.

Sources: SKI 2023 Dalam Angka, BKPK Kemenkes, Tabel 9.5 p. 324 (56.9%) and Tabel 9.7 p. 327 (57.0%); BKPK SKI 2023 oral health factsheet, Gambar 3 p. 2 (11.2%). Product: docs/specs/README.md.

Opportunities

Others answer one part of the question; FindYourDent turns a complaint into a decision and a place Patients can trust

How different is it from any of the other existing ideas?

Alodokter
Health articles, doctor chat and booking. Not a structured dental decision with cost and BPJS for this complaint.

Halodoc
Paid chat with a doctor and medicine delivery. Does not route the complaint to the right kind of dental Provider.

Google Maps
Finds places with open star ratings. No urgency or cost, and reviews are not checked against a visit.

How will it be able to solve the problem?

Urgency tier answers "do I need to go?" for the 57.0% who felt no need.

Cost range and BPJS door before the visit, with why prices vary.

Red-flag check sends emergencies to an IGD before any AI call.

Shortlist of nearby Providers that can treat this complaint.

USP of the proposed solution

"FindYourDent turns a toothache into a guided decision. A grounded Gemini agent explains urgency, the visit journey, cost and BPJS coverage, then shortlists Providers ranked by Patient Testimonies that FindYourDent owns, each one screened by AI and checked against proof of visit."

- **Testimonies FindYourDent owns**
- **Checked against proof of visit**
- **Ranking never paid**

So what The moat is the Testimony data: first-hand, moderated, tied to a visit, and never for sale.

Differentiator: docs/specs/README.md. Competitor scope from public product pages; parity risk analysis in docs/hackathon-plan.md 1.1.

List of features offered by the solution

Eight features cover the Patient journey end to end, and the Provider side starts with a claimed page

1 Guidance card

Urgency, likely category, visit journey, procedures, cost estimate, BPJS and sources.

2 Emergency Escalation

Deterministic red-flag check before Gemini; red “Ke IGD sekarang” card and the nearest IGD.

3 Voice input

Speak the complaint for up to a minute, in Indonesian, a regional language or English.

4 Shortlist

Nearby Providers that can treat the complaint, with reasons, Testimonies and price signals.

5 Provider page

Facts, Testimonies and ratings, WhatsApp and phone; Google rating loads on tap.

6 Moderation

Gemini checks fixed rules; fix-before-submit for phone numbers. Humans remove, AI never deletes.

7 Kunjungan terbukti

A proof of visit marks a Testimony as a proven visit, the defence against buzzers.

8 Claim and Price List

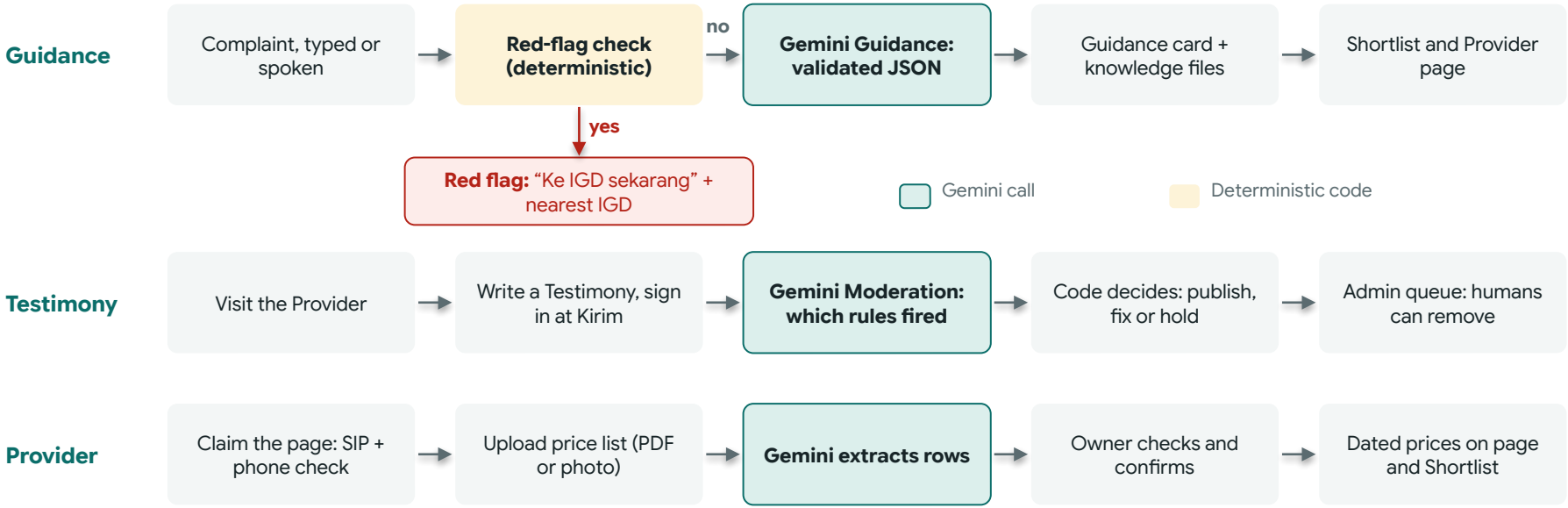
Owner verifies with a SIP; Gemini reads the price list into rows the owner confirms.

Also live: shareable Guidance link without the complaint · Indonesian and English · live impact counter · directory by kota/kabupaten · sign-in only at “Kirim”

Specs 01 to 05 and 09, docs/specs/. Every feature listed is in the deployed build at findyourdent.com.

Process flow diagram or Use-case diagram

Safety runs first: a deterministic red-flag check decides before Gemini sees the complaint



Specs 01, 03 and 05 in docs/specs/; ADR-0004 (AI publishes, humans remove). The model can raise urgency but never lower it below a red flag.

Wireframes/Mock diagrams of the proposed solution

Mobile-first: one screen asks, one answers, one shows where to go



Ask. Free text or voice; disclaimer and no-storage notice up front.

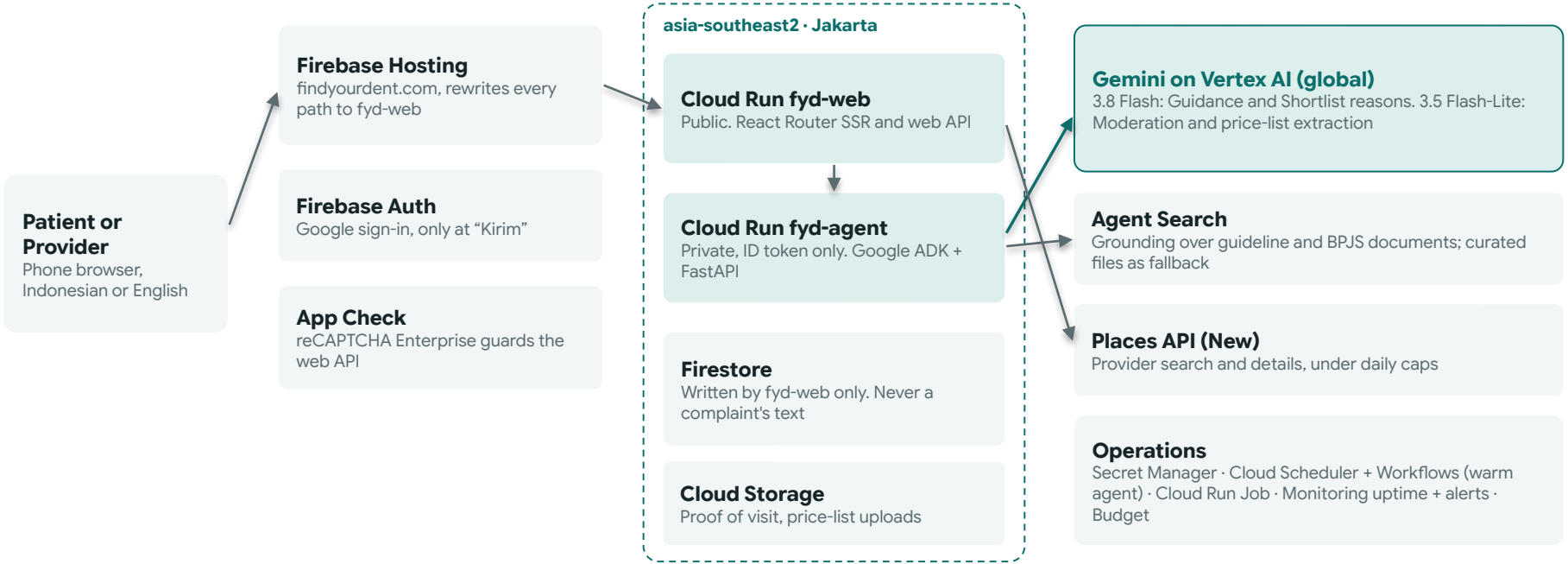
Answer. Urgency first, then problem, visits, cost and BPJS, with sources.

Go. Shortlist cards end in two actions: the Provider page or Google Maps.

Wireframes redrawn from the mockups in docs/design/mockups/html (Beranda, Panduan, Darurat, Tempat), built mobile-first at 360 px.

Architecture diagram of the proposed solution

Two Cloud Run services in Jakarta; only the private agent calls Gemini



Deployed build, 18 October 2026: README.md, infra/gcp/README.md, ADR-0003 (Gemini global endpoint), ADR-0008 (Hosting in front of Cloud Run). The browser never calls the agent, Places or Gemini.

Technologies to be used in the solution

Firebase at the edge, Firestore for data, Cloud Run for compute, Gemini for judgement

Firebase

Hosting serves findyourdent.com and rewrites to Cloud Run. Auth signs Patients and Providers in with Google. App Check with reCAPTCHA Enterprise guards the web API.

Firestore

In Jakarta: Providers, Testimonies, Moderation decisions and owner-confirmed price lists. Never a Patient's complaint text.

Cloud Run

In Jakarta: the public web service, and a private Google ADK agent service that only the web service can call. One warm agent instance during judging.

Gemini

On Vertex AI: writes the Guidance card as validated structured output, moderates every Testimony before it is published, reads price lists into rows the owner confirms.

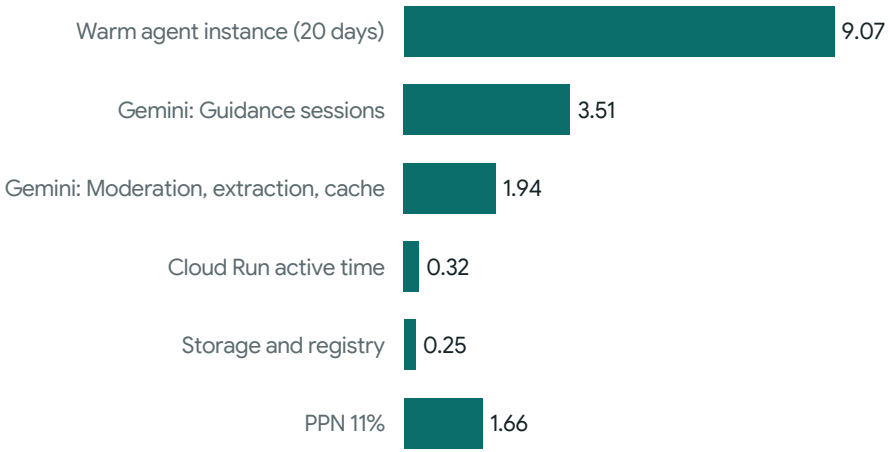
More Google Cloud	Google ADK · Agent Search · Places API (New) · Cloud Storage · Secret Manager · Cloud Scheduler · Workflows · Cloud Monitoring
Web	React Router v7 (SSR) · TypeScript · Tailwind v4 · shadcn/ui · i18next
Agent	Python 3.13 · FastAPI · Pydantic v2 · uv
Contracts and tests	JSON Schema (Ajv + generated Pydantic) · Vitest on Firebase emulators · pytest · Playwright · Biome · Ruff · GitHub Actions

docs/submission/brief-description.md (issue #109), README.md.

Estimated implementation cost

Judging traffic costs about USD 17 for 23 days, and daily quotas cap every paid API

USD, 300 SESSIONS, 16 OCT TO 7 NOV 2026 (COLUMN B2)



Total USD 16.75 with PPN. At 2,000 sessions: USD 37.77. The warm instance is the biggest line; Cloud Run's free tier should cut it.

HARD CAPS SET IN THE PROJECT

Quota	Daily cap
Places Text Search	160 calls
Places Get Place (details, reviews)	83 calls
Places Autocomplete	300 calls
Places Nearby, Photos	0
Gemini 3.8 Flash input tokens	2,000,000
Gemini 3.5 Flash-Lite input tokens	1,000,000
Billing budget, whole account	IDR 1.65 M (~USD 100), alerts 50% and 90%

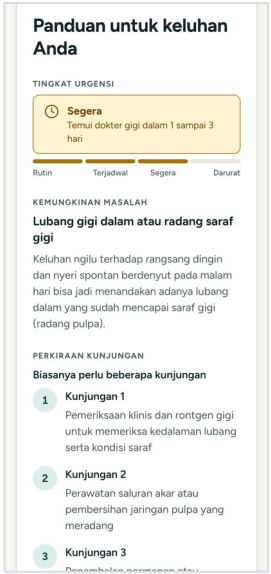
docs/research/cost-comparison-2026-10-17.md (list prices read 16 to 17 Oct 2026, ex-tax plus PPN 11%; no credits); infra/gcp/README.md daily caps. Places spend above free allowances capped at USD 10 a month (ADR-0005).

Snapshots of the prototype

The live build at findyourdent.com, captured at phone width



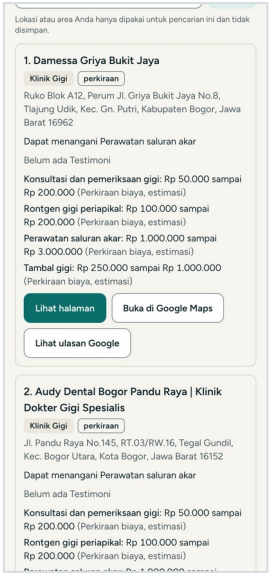
Home: describe the complaint



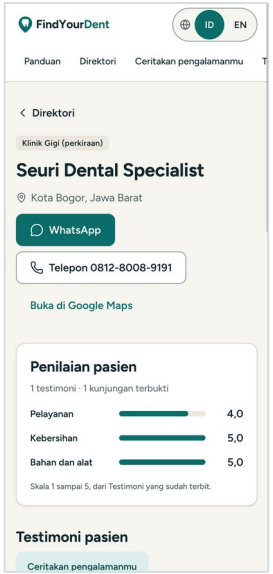
Guidance: urgency, problem, visits



Emergency: “Ke IGD sekarang”



Shortlist: who can treat it nearby



Provider page: Testimonies, ratings

Screenshots of https://findyourdent.com at 390 px (Chrome DevTools, mobile emulation), synthetic complaints. Testimonies shown are real Patient posts published through Moderation.

Prototype Performance report/Benchmarking

Every safety gate passes on the live build; urgency-tier accuracy is the one gate still failing

48/48

must-escalate phrases return “Ke IGD sekarang” without calling Gemini

28/28

ordinary complaints do not escalate (negations, slang, regional languages)

0

unsafe medicine advice across the 40-case Guidance evaluation

0

rule-breaking Testimonies published across 30 cases × 3 runs

Metric (live models, 18 Oct 2026)	Result	Gate
Emergency recall, Guidance	100% (8/8)	**pass**
Urgency-tier accuracy	76% (28/37)	**fail, ≥ 85%**
Category / journey accuracy	100% / 100%	
BPJS accuracy	87% (20/23)	
Moderation decision accuracy	96.7% (29/30)	**pass, ≥ 90%**
Moderation neutrality mismatches	0 pairs	**pass**
Lighthouse mobile: a11y, practices, SEO	100 / 100 / 100	
Lighthouse performance, e2e tests	in progress (T19)	

Honest limits: 8 of the 9 urgency misses are more urgent than the label, the safe direction. Guidance labels are source-derived and agent-labelled, not yet reviewed by a licensed dentist.

README.md: Emergency Escalation results, Guidance evaluation (gemini-3.8-flash, evalset 2026-10-17.1), Moderation evaluation (gemini-3.5-flash-lite). Lighthouse via Chrome DevTools, 18 Oct 2026.

Additional Details/Future Development

Providers and clinics will pay for tools, never for rank

Revenue from Providers and clinics

- Verified-profile subscription with price-list hosting for claimed pages
- Clinic intake: the shared Guidance link becomes the pre-visit note
- Payment Method on Testimonies (spec 07) for price signals

Rules that do not bend

- Ranking is never paid; a claim does not change the Shortlist order
- No ads, promotion, discounts or per-lead fees, in line with KODEKGI and MKEK Fatwa 29/2021
- Providers can reply or report, never delete a Testimony
- Education and navigation, not diagnosis or prescription

Next build

- Lift urgency-tier accuracy past the 85% gate
- A licensed dentist reviews the 40-case evalset
- Proven Testimonies part 2 and buzzer defence (spec 09)
- Kecamatan coverage and more cities

So what Trust is the product: if rank could be bought, the Testimonies would stop meaning anything.

Specs 05, 07 and 09 in docs/specs/; monetisation and ethics constraints in docs/hackathon-plan.md 1.2; Terms and About pages on findyourdent.com.

Provide links to your:

- GitHub Public Repository: https://github.com/ahmadwaliradhi/findyourdent
- Demo Video Link (3 Minutes): https://youtu.be/KsSCkUG_M00
- Final Product Link: https://findyourdent.com

Code

github.com/ahmadwaliradhi/findyourdent

Demo video

youtu.be/KsSCkUG_M00

Live product

findyourdent.com

Each link opened in a logged-out browser before submission. The live demo stays up through the finalist announcement on 7 November 2026.

Sources and notes

SKI 2023 figures	Survei Kesehatan Indonesia 2023 Dalam Angka, BKPK Kemenkes, Bab 9: Tabel 9.5 p. 324, Tabel 9.7 p. 327. badankebijakan.kemkes.go.id/ski-2023-dalam-angka
11.2% treated	BKPK SKI 2023 oral health factsheet, Gambar 3 p. 2. repository.badankebijakan.kemkes.go.id/5534
Reading the 57.0%	Its denominator is everyone aged 3+ not treated by a dental health worker, including people with no dental problem (multiple response).
Cost	docs/research/cost-comparison-2026-10-17.md, USD list prices, ex-tax, PPN 11% as one line, no credits.
Evaluations	README.md results sections; evalsets in services/agent/evals/. Guidance labels agent-labelled, Moderation labels team-labelled.
Ethics	KODEKGI (Kode Etik Kedokteran Gigi Indonesia) and MKEK Fatwa 29/2021 on advertising by doctors and dentists.

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Thank you

